AJAYKRISHNAN E S

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EDUCATION

Ph.D. in Computer Science 2023 - current
University of California, Santa Barbara CGPA - 4/4
Advisor: Prof. Daniel Lokshtanov
Topics: Graph Algorithms, Algorithmic Graph Theory

BS-MS (Specialisation in Mathematics) 2018 - 2023
Indian Institute of Science Education and Research, Pune CGPA - 9.2/10
Thesis Title: Exact Exponential Algorithms & Knot-Free Vertex Deletion

AWARDS AND RECOGNITION

Recipient of **Chuck Dana Research Accelerator Award** at UC Santa Barbara. 2026
Awarded $8,000; sole recipient from a competitive pool of UCSB CS graduate students.

Co-author of the paper that received the **Mihai Pătrașcu Best Paper Award** at SOSA. 2026
Title: A Quasi-Polynomial Time Algorithm for 3-Coloring Circle Graphs.

Recipient of the **Outstanding Teaching Assistant** award at UC Santa Barbara. 2025
Nominated by graduating seniors; awarded to one TA per department annually.

Recipient of UCSB CS **Summer Academic Fellowship**. 2024
Awarded $10,000 to support independent research over the summer.

Recipient of the **Xytel Best Thesis Award** in Mathematics at IISER Pune. 2023
Awarded to one graduating student per discipline each year.

Graduated with distinction from IISER Pune. 2023
An honor reserved for students with a CGPA above 9 (approximately the top 5%).

Recipient of the **Innovation in Science Pursuit for Inspired Research** scholarship. 2018
Five-year fellowship from the Government of India for pursuing higher studies in science.

VISITS AND WORKSHOPS

University of Bergen. Summer 2026
Host: Prof. Fedor Fomin.

SoCal Theory Day at University of California San Diego Feb 2026

Institute of Mathematical Sciences, Chennai. Summer 2025
Host: Prof. Saket Saurabh.

Workshop on Hardness of Approximation in P at DIMACS, Rutgers University July 2025

Fusing Theory and Practice of Graph Algorithms at ICERM, Brown University Feb 2025

Extroverted Sublinear Algorithms at Simons Institute June 2024

Randomized Methods for Approximation and Parameterized Algorithms. Dec 2022

Algorithms - To Improve, or Not to Improve, That's the Question! Jan 2022

SELECT PUBLICATION

Note: In theoretical computer science, authors are listed in alphabetical order. Among my publications at venues which use ranked authorship, those marked with (⋆) indicate equal contribution from all authors.

(Treewidth, Clique)-Boundedness and Poly-logarithmic Tree-Independence.
Maria Chudnovsky, **Ajaykrishnan E S**, Daniel Lokshtanov. *Accepted - IJM*

Parameterized Approximation of Rectangle Stabbing.
Huairui Chu, **Ajaykrishnan E S**, Daniel Lokshtanov, Anikait Mundhra, *ESA 2026*
Thomas Schiber, Xiaoyang Xu, Jie Xue.

A Quasi-Polynomial Time Algorithm for 3-Coloring Circle Graphs. ***(Best Paper Award)***
Ajaykrishnan E S, Robert Ganian, Daniel Lokshtanov, Vaishali Surianarayanan. *SOSA 2026*

Fast Hypertree Decompositions via Linear Programming:. **(⋆)**
Fractional and Generalized
Vaishali Surianarayanan, Anikait Mundhra, **Ajaykrishnan E S**, Daniel Lokshtanov. *SIGMOD 2025*

Beyond Exact Fairness: Envy-Free Incomplete Connected Fair Division.
Ajaykrishnan E S, Daniel Lokshtanov. *FSTTCS 2025*

UNDER SUBMISSION

Tree independence IX. A Trichotomy for Excluded Induced Minors. *Submitted - JCTB*
Maria Chudnovsky, Julien Codsi, **Ajaykrishnan E S**, Daniel Lokshtanov.

Algorithms for Min-Color Path with Bounded Color-Degree. *Submitted - STOC*
Ajaykrishnan E S, Daniel Lokshtanov, Saket Saurabh, Jie Xue.

Induced Minors and Coarse Tree Decompositions. *Submitted - JGT*
Maria Chudnovsky, Julien Codsi, **Ajaykrishnan E S**, Daniel Lokshtanov.

Computing Join Trees for Join Queries with Inequalities. *Submitted - ICDT*
Ajaykrishnan E S, Daniel Lokshtanov, Anikait Mundhra,
Vaishali Surianarayanan, Nikolaos Tziavelis.

Tree-independence number VIII. 3PC-free graphs have logarithmic *Submitted - DM*
tree-independence number.
Maria Chudnovsky, Julien Codsi, **Ajaykrishnan E S**, Daniel Lokshtanov, Ilya Maier.

SERVICE

Conference Reviewing:

FOCS 2024, ITCS 2025, COCOON 2025, ESA 2025, STACS 2026, SODA 2026, IWOCA 2026, MFCS 2026, IPEC 2026, SODA 2027, SOSA 2027, SOFSEM 2027.

Mentorship Roles:

UC Santa Barbara, Early Research Scholars Program; a research program that seeks to give students, *particularly those who are traditionally under-served* in computing the foundational knowledge and skills for engaging in research in the discipline.

- Mentored a team of three undergraduate students. 2024-2025
  Project Title: Teaching Assistant Assignment.
- Mentored a team of four undergraduate students. 2025-2026
  Project Title: Low Distortion Metric Embedding.

Teaching Experience:

- Teaching Assistant at UCSB for CMPSC 130B - Data Structures and Algorithms II
  With Prof. Subhash Suri; Fall 2023, Spring 2024, Winter 2025.
- Teaching Assistant at UCSB for CMPSC 130B - Data Structures and Algorithms II
  With Prof. Eric Vigoda; Winter 2024.
- Teaching Assistant at UCSB for CMPSC 130B - Data Structures and Algorithms II
  With Prof. Daniel Lokshtanov; Fall 2024.
- Teaching Assistant at UCSB for CMPSC 134 - Randomized Algorithms
  With Prof. Eric Vigoda; Spring 2025.
- Teaching Assistant at UCSB for CMPSC 40 - Foundations of Computer Science
  With Dr. Vaishali Surianarayanan; Summer 2024.
- Teaching Assistant at IISER-Pune for MTH 323 - Graph Theory
  With Prof. Soumen Maity; Fall 2022.